

Activity 6

Aim: To determine pH of HCl solution of various concentrations by using pH paper or universal indicator.

Apparatus: Test tubes, test tube stand.

Chemicals: 0.1M, 0.01M, 0.001M HCl solution, pH paper or universal indicator.

Theory: $\text{pH} = -\log_{10}[\text{H}^+]$

For Acidic solution, $\text{pH} < 7$, Basic solution $\text{pH} > 7$, and neutral solution $\text{pH} = 7$.

The value of pH changes with change in concentration of HCl.

Procedure:

1. Take 5 ml each of 0.1 M, 0.01 M, 0.001 M HCl solutions in separate test tubes marked A, B, C.
2. Add 2 drops of universal indicator in each test tube or put piece of pH paper in each test tube.
3. Observe the Color of solution or pH paper and record pH of each solution.

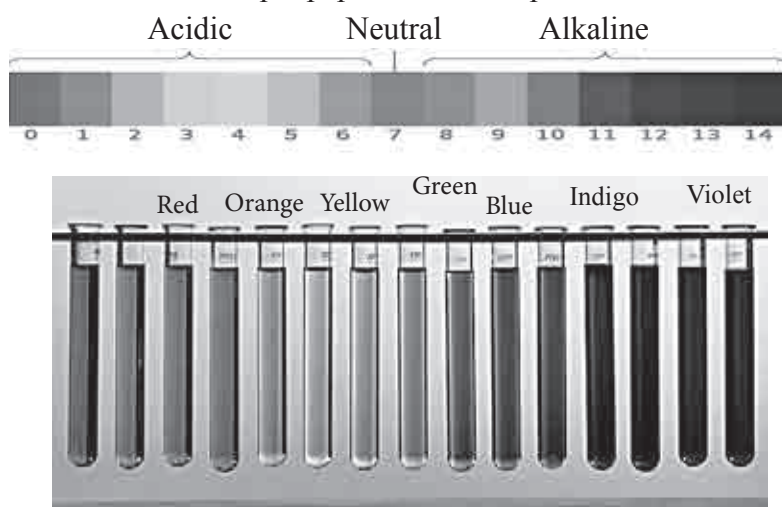


Fig. Activity 6.1 pH range

Note: Student has to prepare 0.01M and 0.001M HCl solution from the given 0.1M HCl solution.

Observation Table/ Conclusion:

Sr. No.	Solutions	Concentration of solutions	Color of pH paper or solution	pH
1	A	0.1 M	Red	1
2	B	0.01 M	Light red	2
3	C	0.001 M	Orange	3

Remark and sign of teacher:

MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

1. If nature of the solution is acidic, then its pH value is ---
a. ☒ > 7 c. equal to 7
b. ☐ < 7 d. in between 7 to 14
2. Which of the following indicator have different pH range is --
a. phenolphthalein indicator c. methyl orange indicator
b. acid-base indicator ☒ d. universal indicator
3. The negative log of H^+ ions concentration to the base 10, is known as ---
☒ a. pH c. pKa
b. pOH d. pKb
4. The Color of universal indicator in strongly acidic solution is ----
☒ a. red c. oranges
b. violet d. blue
5. If concentration of H^+ ions increase, the values of the pH of the solution will be -----
☒ a. decrease c. increase
b. remain constant d. first increases then decreases

Short answer questions

Q1. What do you meant by pH of the solution?

Ans. The pH of a solution is a measure of the molar concentration of hydrogen ions in the solution and hence indicates its acidity or basicity. pH stands for "power of hydrogen".

Q2. If the concentration of given solution of HCl is 0.01M. Calculate the pH of the solution.

Ans. For a strong monoprotic acid like HCl, the hydronium ion concentration is equal to the acid concentration. $[H^+] = 0.01 \text{ M}$ $pH = -\log(0.01)$ $pH = 2$

Remark and sign of teacher:

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